

Mini Study Guide 1: Real vs. Financial Assets

FIN 4000 Investments · Midterm prep · BKM Ch. 1

Learning objectives

A society's wealth comes from its real assets; financial assets are claims on those real assets and the income they produce. The midterm tests this distinction directly with classification questions and indirectly in the Ch. 2 instrument and Ch. 4 fund material.

After this guide you should be able to:

1. Define real and financial assets, and classify any item as one or the other.
2. Explain why financial assets are wealth to their holder but net to zero for the economy as a whole.
3. Trace how a transaction (a loan, an equity issue, a capital purchase) creates or changes each kind of asset.
4. Explain how financial assets help productive capacity indirectly, even though they are not productive capacity themselves.

*Intangible assets (patents, brands, software, data) are covered in the separate intangibles guide. Here they matter only as a category: BKM counts knowledge used to produce goods and services as a **real** asset.*

Core concept

Real assets produce goods and services: land, buildings, machinery, and the knowledge used to run them. **Financial assets** (securities) are claims on real assets or on the income they generate. A share of stock is a claim on a firm's factories and profits. A bond is a claim on promised interest and principal.

Feature	Real assets	Financial assets
What it is	Productive capacity	A claim on productive capacity or its income
Examples	Land, factories, equipment, oil reserves, patents, software, workforce know-how	Stocks, bonds, bank loans, deposits, mutual fund shares, derivatives
Creates wealth?	Yes: it produces output	Not directly: it allocates the output of real assets
Aggregate balance sheet	Remains as national wealth	Nets to zero, because every holder's asset is an issuer's liability
How it is created	Investment: building, buying, or developing capacity	Contracts: issuing securities, making loans
How it is destroyed	Depreciation, obsolescence, disaster	Repayment, default, maturity, contract expiration

The key test: ask who is on the other side. A financial asset always has a counterparty that owes you something (the issuer). A real asset has no counterparty; it produces value on its own.

Why financial assets still matter. They add to productive capacity *indirectly*. They:

- **Separate ownership from management**, so millions of shareholders can own a firm that professional managers run.
- **Move funds** from savers to firms with productive investment opportunities.

- **Transfer and price risk**, so investors bear only the risks they choose.
- **Allow consumption timing**, so savers can shift income from high-earning years to retirement.

Balance-sheet view: why financial assets net out

For one household, financial assets are real wealth. For the whole economy they cancel, and national wealth equals the stock of real assets.

Every financial asset appears twice, once as an asset of the holder and once as a liability or equity claim of the issuer:

$$\text{Net worth}_{\text{economy}} = \sum \text{Real assets} + \sum \text{Financial assets} - \sum \text{Financial liabilities} = \sum \text{Real assets}$$

Where:

- Net worth (economy) = the total wealth of the economy
- Σ Real assets = the sum of all land, buildings, equipment, inventories, and intellectual property
- Σ Financial assets = the sum of all claims (stocks, bonds, loans, deposits) held as assets
- Σ Financial liabilities = the sum of the same claims as owed by their issuers; always equal to Σ Financial assets

Household view vs. economy view. BKM's household balance sheet (Table 1.1) shows most household assets are *financial*: deposits, pension reserves, corporate equity, mutual fund shares. The domestic net worth table (Table 1.2) shows the economy's net worth is all *real*: real estate, equipment, intellectual property, inventories. Both are correct. The household tables list claims; the national table lists what the claims are claims *on*.

Transactions to watch for:

Transaction	Real assets	Financial assets	Society richer?
Firm borrows \$1M from a bank	No change	+\$1M loan (bank's asset) offset by +\$1M debt (firm's liability)	No
Firm uses the \$1M to build a plant	+\$1M plant	Firm's cash falls \$1M	Yes, capacity grew
Firm issues \$5M of new stock for cash	No change	+\$5M shares held by investors, +\$5M cash at firm	No, until the cash is invested
Investor sells stock to another investor	No change	Ownership transfers; no new claim	No
Plant destroyed by fire (uninsured)	– plant	Shares and bonds fall in value to reflect it	Poorer

Worked examples

Example 1: Classify each item (Conceptual)

Keweenaw Kayak Co. lists these items. Which are real assets and which are financial?

Item	Classification	Why
Kayak-molding machine	Real	Produces output

Item	Classification	Why
Warehouse and land in Houghton	Real	Produces services (storage)
Patent on a hull design	Real (intangible)	Knowledge used in production
\$200,000 in a checking account	Financial	Claim on the bank
3-month Treasury bills	Financial	Claim on the U.S. Treasury
Accounts receivable from retailers	Financial	Claim on customers
Inventory of finished kayaks	Real	Goods produced, ready for sale
Call options on a supplier's stock	Financial	Derivative claim whose value depends on another claim

Example 2: A mini economy (Algorithmic)

The economy has three parties:

- **Household A** owns a \$200,000 house and a \$50,000 bond issued by Firm B.
- **Firm B** owns a \$500,000 factory, financed with the \$50,000 bond and \$450,000 of stock.
- **Household C** owns all of Firm B's stock.

Party	Real assets	Financial assets	Liabilities / equity issued	Net worth
Household A	\$200,000	\$50,000	\$0	\$250,000
Firm B	\$500,000	\$0	\$500,000	\$0
Household C	\$0	\$450,000	\$0	\$450,000
Economy	\$700,000	\$500,000	\$500,000	\$700,000

Households report \$700,000 of net worth, of which \$500,000 is financial. National wealth is still \$700,000: the house plus the factory. The \$500,000 of financial claims nets to zero against Firm B's obligations.

Example 3: Tracing transactions (Algorithmic, BKM end-of-chapter style)

Copper Harbor Software starts with \$10,000 of cash from its founders.

1. It borrows \$50,000 from a bank.
2. It spends \$40,000 developing a new software product.
3. It sells the product to a larger firm for 1,500 shares of that firm's stock, worth \$80 per share (\$120,000).
4. It sells 1,000 of those shares at \$80 and repays the bank loan.

After step	Cash	Software (real)	Shares held	Total assets	Loan	Equity	Real / total assets
Start	\$10,000	—	—	\$10,000	—	\$10,000	0%
1. Borrow	\$60,000	—	—	\$60,000	\$50,000	\$10,000	0%
2. Develop	\$20,000	\$40,000	—	\$60,000	\$50,000	\$10,000	66.7%

After step	Cash	Software (real)	Shares held	Total assets	Loan	Equity	Real / total assets
3. Sell product	\$20,000	—	\$120,000	\$140,000	\$50,000	\$90,000	0%
4. Sell shares, repay	\$50,000	—	\$40,000	\$90,000	—	\$90,000	0%

What to notice:

- Step 1 creates a financial asset (the bank's loan) and an equal liability. No real asset is created.
- Step 2 is the only step that creates a real asset.
- Step 3 swaps a real asset for a financial asset. The software still exists; it now sits on the buyer's books. The \$80,000 gain is a revaluation, not new capacity.
- Step 4 swaps one financial asset for another and removes a claim (the loan).

Common pitfalls

- **"Cash is a real asset."** It is not. Currency and deposits are claims on the government or a bank.
- **"Intangible means financial."** It does not. Patents, software, and brands are real assets because they produce output. The test is counterparty, not physical form.
- **"Buying stock is investing."** In the economic sense, buying shares from another investor only moves a claim. Society invests when real capacity is built.
- **"A rising stock market makes the country richer."** Higher prices raise holders' paper wealth. Real wealth rises only if they reflect more or better real assets.
- **Forgetting the other side.** Every loan is a financial asset to one party and a liability to another. Count both.

Key terms and definitions

- **Real assets:** land, buildings, machines, and knowledge used to produce goods and services.
- **Financial assets (securities):** claims on real assets or on the income they generate.
- **Intangible (intellectual property) assets:** real assets without physical form, such as patents, software, and brands. BKM counts them as real assets.
- **Claim / counterparty:** a claim is a right to receive payment; the counterparty is the party obligated to pay it.
- **Fixed-income (debt) securities:** securities that promise a fixed income stream or one set by a formula, such as bonds and T-bills.
- **Equity:** an ownership share in a corporation, with a residual claim and no promised payment.
- **Derivative securities:** securities whose payoffs depend on the prices of other assets, such as options and futures.
- **National wealth vs. household net worth:** national wealth equals the economy's real assets; household net worth also counts financial claims, which net out at the national level.
- **Separation of ownership and management:** shareholders own the firm while professional managers run it, made possible by tradable shares.
- **Consumption timing:** using financial assets to shift purchasing power from high-earning periods to low-earning periods.
- **Allocation of risk:** financial markets let each investor choose how much risk to bear, for example bonds for less risk and stocks for more.

CFA-style practice questions

Q1 [Conceptual]. Which of the following is a real asset?

- A. A share of Ford Motor Company stock
- B. A Ford assembly plant
- C. A car loan made by Ford Credit

Q2 [Conceptual]. A patent held by a pharmaceutical company is *best* classified as:

- A. a financial asset, because it has no physical form.
- B. a real asset, because it is knowledge used to produce goods.
- C. a derivative, because its value depends on future drug sales.

Q3 [Algorithmic]. Household X owns a \$300,000 home and \$100,000 of bonds issued by Firm Y. Firm Y owns \$400,000 of equipment, financed by those bonds and \$300,000 of stock held by Household Z. National wealth is *closest* to:

- A. \$700,000
- B. \$800,000
- C. \$1,100,000

Q4 [Conceptual]. A firm borrows \$2 million from a bank and leaves the proceeds on deposit. Immediately after the loan:

- A. the economy's real assets increase by \$2 million.
- B. financial assets and financial liabilities each increase by \$2 million.
- C. national wealth increases by \$2 million.

Q5 [Conceptual]. Financial assets contribute to an economy's productive capacity mainly by:

- A. producing goods and services directly.
- B. channeling savings to firms and allowing ownership to be separated from management.
- C. adding to national wealth whenever security prices rise.

Q6 [Algorithmic]. A firm holds \$30,000 in cash, \$70,000 of equipment, a patent valued at \$20,000, and \$40,000 of Treasury bills. The proportion of its total assets that are real assets is *closest* to:

- A. 43.8%
- B. 56.3%
- C. 75.0%

Q7 [Conceptual]. Someone discovers a buried chest holding \$10 billion in U.S. currency. Which statement is *most* accurate?

- A. Society is \$10 billion richer in real terms.
- B. The currency is a financial asset. The finder is richer, but society's real productive capacity is unchanged.
- C. The currency is a real asset because it is physical.

Q8 [Conceptual]. Which transaction increases the economy's real assets?

- A. A pension fund buys existing shares on the NYSE.
- B. A utility builds a new wind farm.
- C. A bank makes a mortgage loan to the buyer of an existing home.

Answer key

Q	Answer	Explanation
1	B	The plant produces cars. Stock (A) and a loan (C) are claims.

Q	Answer	Explanation
2	B	BKM counts knowledge used in production as a real asset. Physical form does not decide it (A). A patent is not a contract whose value comes from another security (C).
3	A	National wealth = real assets = \$300,000 home + \$400,000 equipment = \$700,000. The \$400,000 of bonds and stock nets out. C wrongly adds the financial claims.
4	B	The bank's loan asset and the firm's liability are equal and offsetting. The firm's deposit is also matched by the bank's deposit liability. No real capacity exists until the funds are spent on real assets.
5	B	Financial assets help indirectly by moving savings to investment and allowing separation of ownership and management. C confuses paper wealth with real wealth.
6	B	Real = \$70,000 + \$20,000 = \$90,000. Total = \$160,000. $90 / 160 = 56.25\%$. A omits the patent; C counts cash as real.
7	B	Currency is a claim (a government liability). The finder gains purchasing power, but no new productive capacity exists.
8	B	Only the wind farm adds productive capacity. A and C transfer or create claims on existing assets.